

LL(1) - Results



Attempt 1 of 1

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Attempt Score **10 / 10 - 100 %**

Overall Grade (Highest Attempt) **10 / 10 - 100 %**

Question 1

1 / 1 point

Consider the grammar:

$S \rightarrow A B$

$A \rightarrow a \mid \epsilon$

$B \rightarrow b$

What is $\text{FIRST}(A)$?

- ☐ {a}
- ☐ {a, b}
- ☐ { ϵ }
- ☒ {a, ϵ }

Question 2

1 / 1 point

What is $\text{FIRST}(S)$ for the same grammar?

- ☐ {a}
- ✓ ☒ {a, b}
- ☐ {b}
- ☐ {a, ϵ }

Question 3

1 / 1 point

What is FOLLOW(A)?

- ☐ {a}
- ✓ ☒ {b}
- ☐ {\$}
- ☐ {a, b}

Question 4

1 / 1 point

Consider the grammar:

$S \rightarrow A B$

$A \rightarrow a \mid \epsilon$

$B \rightarrow b \mid \epsilon$

What is FIRST(S)?

- ☐ {a}
- ☐ {a, b}

✓ ☒ {a, b, ϵ }

☐ { ϵ }

Question 5

1 / 1 point

What is FOLLOW(A)?

☐ {b}

✓ ☒ {b, \$}

☐ {\$}

☐ {a}

Question 6

1 / 1 point

Which grammar cannot be parsed by a predictive top-down parser without modification?

☐ $S \rightarrow aA$
 $A \rightarrow b$

☐ $S \rightarrow Aa$
 $A \rightarrow b \mid \epsilon$

✓ ☒ $S \rightarrow Sa \mid b$

☐ $S \rightarrow aSb \mid \epsilon$

Question 7

1 / 1 point

What does LL(1) mean?

- ✓ ☒ Left-to-right scan, Leftmost derivation, 1 lookahead
- ☐ Left recursion allowed, 1 stack
- ☐ Leftmost derivation, LR parser
- ☐ Linear language

Question 8

1 / 1 point

Which grammar transformation preserves language?

- ☐ Left recursion elimination
- ☐ Left factoring
- ☐ Removing ambiguity
- ✓ ☒ All of the above

Question 9

1 / 1 point

Given grammar

$S \rightarrow A B$

$A \rightarrow a \mid \epsilon$

$B \rightarrow b$

What is **FIRST(AB)**?

- ☐ {a}

- ☐ {b}
- ✓ ☒ {a, b}
- ☐ { ϵ }

Question 10

1 / 1 point

Given the grammar

$S \rightarrow A a$

$A \rightarrow a \mid \epsilon$

Is this grammar **LL(1)**?

- ☐ yes
- ✓ ☒ no
- ☐ not enough information to tell

Done